

ENSF 380

TOPIC 03: JAVA PACKAGES

Multiple Files

01_SamePackage

OneClass.java

```
public class OneClass {  
    public static void main(String args[])  
    {  
        TwoClass object = new TwoClass(2);  
        object.output();  
    }  
}
```

TwoClass.java

```
public class TwoClass {  
    private int value;  
  
    public TwoClass(int number) {  
        this.value = number;  
    }  
    public void output() {  
        System.out.println("TwoClass was "  
        + "instantiated with " + value);  
    }  
}
```

Multiple Files in Different Packages

02_DifferentPackage

OneClass.java

```
import apackage.TwoClass;

public class OneClass {
    public static void main(String args[])
    {
        TwoClass object = new TwoClass(2);
        object.output();
    }
}
```

apackage/TwoClass.java

```
package apackage;

public class TwoClass {
    private int value;

    public TwoClass(int number) {
        this.value = number;
    }

    public void output() {
        System.out.println("TwoClass was "
            + "instantiated with " + value);
    }
}
```

Fully Qualified Package

03_FullyQualified

OneClass.java

```
import edu.ucalgary.oop.*;

public class OneClass {
    public static void main(String args[])
    {
        TwoClass object = new TwoClass(2);
        object.output();
    }
}
```

edu.ucalgary.oop/TwoClass.java

```
package edu.ucalgary.oop;

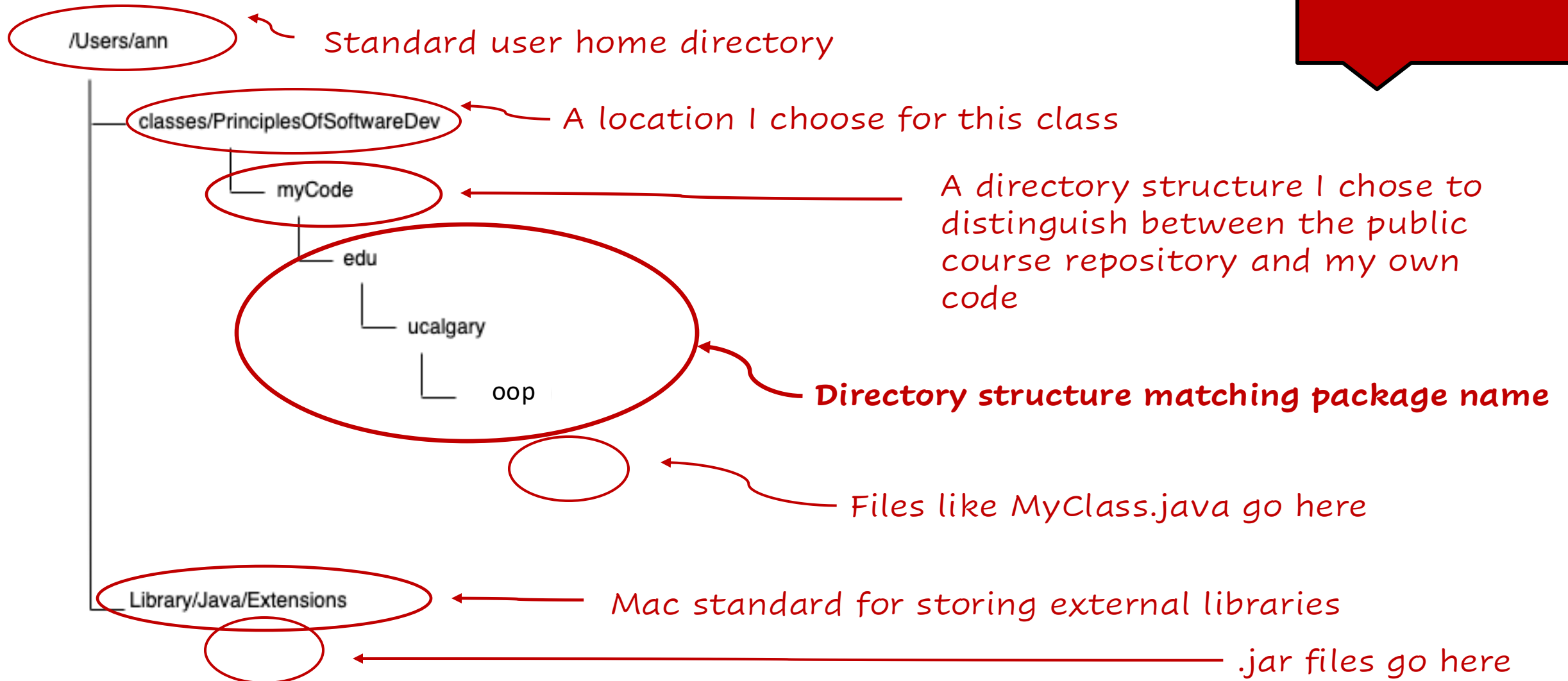
public class TwoClass {
    private int value;

    public TwoClass(int number) {
        this.value = number;
    }

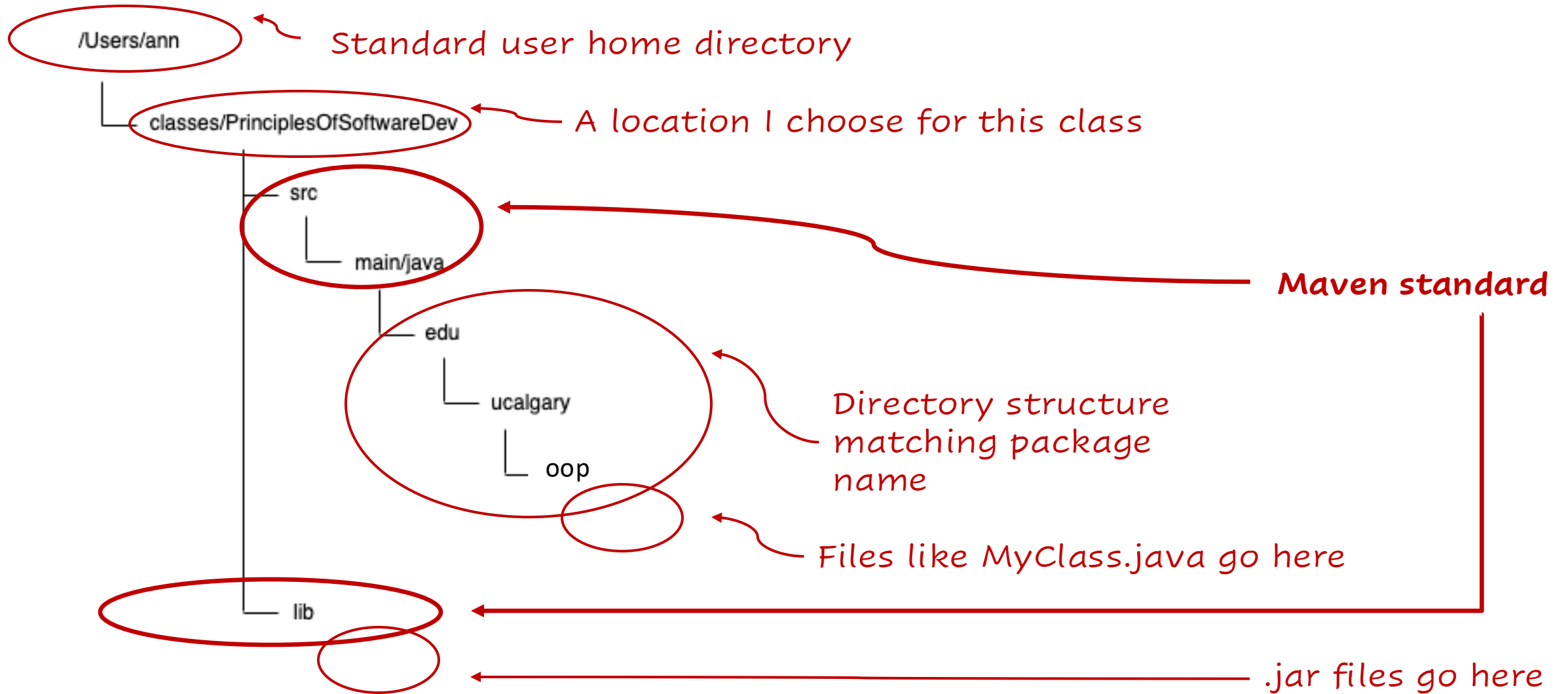
    public void output() {
        System.out.println("TwoClass was "
            + "instantiated with " + value);
    }
}
```

Understanding Java File Locations

Important: The edu/ucalgary/oop portion of the directory structure is necessary.



Understanding Java File Locations



Compiling and Running: Command-Line

03_FullyQualified

1. From the directory outside the package, compile with the complete path.
 - If TwoClass.java is located in: `/Users/home/ann/code/main/edu/ucalgary/oop/`
 - From the directory: `/Users/home/ann/code/main`
 - Type: `javac edu/ucalgary/oop/TwoClass.java`
2. From the directory outside the package, run with the complete path.
 - If TwoClass.java is located in: `/Users/home/ann/code/main/edu/ucalgary/oop/`
 - From the directory: `/Users/home/ann/code/main`
 - Type: `java edu.ucalgary.oop.TwoClass`

DEMO

COMPILING AND RUNNING FROM THE COMMAND LINE

Sub Packages

04_SubPackage

edu/ucalgary/oop/subclass/ThreeClass.java

```
package edu.ucalgary.oop.subclass;

public class ThreeClass {
    private int value;

    public ThreeClass(int number) {
        this.value = number;
    }
    public void output() {
        System.out.println("ThreeClass was "
            + "instantiated with " + value);
    }
}
```

edu/ucalgary/oop/TwoClass.java

```
package edu.ucalgary.oop;

public class TwoClass {
    private int value;

    public TwoClass(int number) {
        this.value = number;
    }
    public void output() {
        System.out.println("TwoClass was "
            + "instantiated with " + value);
    }
}
```

Packages

Used to group together related classes

Written in lower case letters

Path indicates fully qualified name

Often embedded in an additional directory structure

- e.g., Maven uses `src/main/java`

Advantages to Packages

Reusability

Organization

Avoid name conflicts

Name Conflicts

05_NameConflict

OneClass.java

```
import edu.ucalgary.oop.*;
import edu.ucalgary.oop.layer.*;

public class OneClass {
    public static void main(String args[]) {
        var one = new
            edu.ucalgary.oop.TwoClass(2);
        var two = new
            edu.ucalgary.oop.layer.TwoClass(200);
    }
}
```

edu/ucalgary/oop/TwoClass.java

```
package edu.ucalgary.oop;

public class TwoClass {
    ...
}
```

edu/ucalgary/oop/layer/TwoClass.java

```
package edu.ucalgary.oop.layer;

public class TwoClass {
    ...
}
```

Course Expectations

Files and code

- Only one class in each .java file (except helper classes)
- **Each class will be part of the package `edu.ucalgary.oop`**
- **Each file must contain a fully qualified package name**
- If you use a `main()` method for development, it should be in a separate class (from the one(s) requested)
- Classes need to be public to be accessible

Submission format

- The submitted file is `12345678.zip` where 12345678 corresponds to your student ID
- For group submissions, use the group ID, e.g., `40.zip` for group 40

Note

- Do not use subpackages nested within `edu/ucalgary/oop`
- Directory structure of submissions may **optionally** include the package structure (or continue as before)
 - `12345678/edu/ucalgary/oop/Class.java` (allowed)
 - `12345678/Class.java` (allowed)

Summary: Knowledge

Why packages are used

The relationship between directory structure and fully qualified name

Organizing your Java directories

Summary: Skills

How to create a package in Java

How to use a package in Java

Additional Information

Related reading

- Volume 1, Chapter 4.7: Packages

Course expectations

- **Failure to use the fully qualified package name will result in a compilation failure in the test environment.**